INCH-POUND

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DETAIL SPECIFICATION

V-22 CORD, DETONATING ASSEMBLY

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWCIHD) and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers V-22 detonating cord assemblies (DCA).

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

MIL-PRF-46010 - Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331 - Fuzes, Ignition Safety Devices and Other Related Components, Environmental and Performance Tests for

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 1377

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NSWC IHD SPECIFICATIONS (CAGE CODE 14083)

WS 5003 - HNS Explosive

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

3205AS109 - Cord, Detonating Assembly, Primer/High Explosive

3205AS110 - Cord, Detonating Assembly, High Explosive/Low Energy

3205AS111 - Cord, Detonating Assembly, High Explosive/Gas Generator

3205AS182 - Firing Fixture Assembly

3205AS217 - Closed Bomb Fixture

(Copies of these publications are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil).

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of this publication are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.3), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design and construction. The DCAs and all its components shall be manufactured in accordance with NAVAIR Drawings 3205AS109, 3205AS110, and 3205AS111, and the part numbers specified in table I. The design and construction of the DCAs shall meet all the requirements of this specification to ensure inherent reliability so that the equipment will consistently perform as specified. The requirements of MIL-D-21625 shall apply as requirements of this specification for manufacturing, testing, and procurement with the exceptions and additions specified herein.

TABLE I. Part numbers.

NAVAIR Drawing no.	**Part numbers**					
3205AS109	-1	-2	-3	-4	-5	-6
3205AS110	N/A	-2	N/A	N/A	N/A	N/A
3205AS111	-1	N/A	N/A	N/A	N/A	N/A

3.2.1 End fitting torque. Aluminum end fittings shall withstand 90 inch-pounds minimum assembly torque without deformation and stainless steel end fittings shall withstand 120 inch-pounds.

3.2.2 Energy stimulus. The DCAs shall be configured such that the energy stimulus shall be contained and shall not emit shrapnel when fired in accordance with 4.5.13 and in the “as-installed” condition.

3.2.3 Intact. The DCA end fittings shall remain intact in the port. Out-gassing of the DCA and fraying of the stainless steel overbraid are permissible, provided that some of the overbraid remains intact and the aforementioned

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criteria is met. DCAs shall be constructed such that the portion of the assembly between the end fittings shall not separate or disengage from the end fitting when fired in accordance with 4.5.13 and in the "as-installed" condition over the specified temperature range specified herein.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform to the standards established at technical service release as specified (see 6.2). Any divergence shall require technical evaluation and potential design certification testing.

3.3.1 Primary components. For the purpose of this specification, the stabilized, braided, and vibrated thin layer explosive (TLX) cord, the percussion primer, the loaded booster shell (HNS,WS 5003), and the ferrule check valve assembly are considered primary components. Only one lot of each primary component shall be used in a DCA lot. One primary lot may be used in more than one DCA lot.

3.3.1.1 Percussion primers. Only M42C1-979 percussion primers or equal shall be used. Only primers manufactured within the last 24 months or recertified within the last 12 months shall be used in the DCAs.

3.4 DCA requirements.

3.4.1 Visual requirements. The DCA shall be free of the following defects: illegible, missing, or inaccurate markings, damage, burrs, dents, sharp edges, chipped or incomplete protective finish, or other defects that may prevent the installation of the DCA when examined in accordance with 4.5.4. Each DCA shall meet the requirements of NAVAIR Drawings 3205AS109, 3205AS110, and 3205AS111.

3.4.2 Radiographic requirements.

3.4.2.1 X-ray. X-ray shall be conducted in accordance with 4.5.5.1. When radiographically examined in accordance with 4.5.5, the DCA shall show proper assembly, presence of parts, uniformity of propellant, and sealing. In particular, the x-rays shall be checked for proper orientation of the primer anvil within the primer cup, and the presence of the check ball in the gas generator portion of that DCA. The radiographic inspection shall show no foreign materials present.

3.4.2.2 N-ray. N-ray shall be conducted in accordance with 4.5.5.2. When radiographically examined in accordance with 4.5.5, the DCA shall show the proper assembly, presence of parts, uniformity of propellants, and sealing. The radiographic examination shall show no foreign materials present.

3.4.3 End fittings leakage. End fittings shall exhibit a leak rate of not greater than 1×10^{5} cubic centimeter per second (cc/sec) of dry air when tested in accordance with 4.5.6.

3.4.4 Detonation velocity. The cord detonation velocity shall be 1,400 to 2,500 meters/second when tested in accordance with 4.5.1.

3.4.5 Corrosion. All components used in production shall be free of corrosive products in accordance with 4.5.2. The threads of the DCA end fittings shall be coated with a dry film lubricant that meets the requirements of MIL-PRF-46010.

3.4.6 Explosive weight. The explosive weight of the cord shall be 20±10 milligrams of the HMX and aluminum mixture per meter when tested in accordance with 4.5.3.

3.5 Environmental requirements.

3.5.1 Vibration. The DCA shall not initiate or incur any internal or external damage or degradation when vibration cycles are applied in accordance with 4.5.7.

3.5.2 Shock. The DCA shall not initiate or incur any internal or external damage or degradation when sawtooth shocks are applied in accordance with 4.5.8.

3.5.3 Temperature-shock/humidity/altitude (TSHA) cycling. The DCA shall not initiate or incur any internal or external damage or degradation when subjected to TSHA cycling in accordance with 4.5.9.

3.5.4 Salt fog. The DCA shall not initiate or incur any internal or external damage or degradation when subjected to salt fog testing in accordance with 4.5.10.

3.5.5 Six-foot drop. The DCA shall not initiate or incur any major internal or external damage or degradation when subjected to the 6-foot drop test in accordance with 4.5.11.

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3.5.6 Forty-foot drop. The DCA shall not initiate or incur any internal or external damage that makes the DCA unsafe for disposal when subjected to the 40-foot drop in accordance with 4.5.12.

3.6 Functional requirements. The DCA shall meet the requirements of 3.6.1, 3.6.2, or 3.6.3 when fired in accordance with 4.5.13.

3.6.1 Primer-high. The DCA shall be initiated by energizing a solenoid firing mechanism that is in accordance with the fixture shown in NAVAIR Drawing 3205AS182. The DCA shall initiate and propagate along its complete length. The output end shall dent a 6061-T6 aluminum dent block conditioned to 70±5 °F when the DCA is at the following conditions:

a. At -65 °F, a minimum indent depth of 0.035 inch.

b. At +70 °F, a minimum indent depth of 0.040 inch.

c. At +200 °F, a minimum indent depth of 0.040 inch.

3.6.2 High-low. The DCA shall initiate along its complete length. The high order output end shall dent a 6061-T6 Aluminum dent block conditioned to 70±5 °F when the DCA is at the following conditions:

a. At -65 °F, a minimum indent depth of 0.035 inch.

b. At +70 °F, a minimum indent depth of 0.040 inch.

c. At +200 °F, a minimum indent depth of 0.040 inch.

When reversed fired, initiating the high order end, the low order (energy) end shall not initiate a joined high order end.

3.6.3 High-gas. When initiated by an SMDC or TLX test tip, the DCA shall propagate and initiate the gas generator tip. At all temperatures, the gas generator shall produce a minimum pressure of 200 pounds per square inch gauge (psig) within 10 milliseconds in a 10.5 cubic centimeter closed bomb that is in accordance with NAVAIR Drawing 3205AS217. The pressure remaining after 20 milliseconds shall be greater than 200 psig. The peak pressure shall not exceed 3,000 psig for a duration greater than 200 microseconds.

3.7 Workmanship. The DCA shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. The DCA shall be free of the following defects: illegible, missing, or inaccurate markings, damage, burrs, dents, sharp edges, chipped or incomplete protective finish, or other defects which may prevent the installation of the DCA. Each DCA shall meet the requirements of their respective NAVAIR Drawing (3205AS109, 3205AS110, and 3205AS111).

4. VERIFICATION

4.1 Classification of inspections.

a. First article inspection (see 4.2).

b. Production inspection (see 4.3).

c. Lot acceptance inspection (see 4.4).

4.2 First article inspection. Unless otherwise specified (see 6.2 and 6.3), a first article inspection sample of 33 DCAs in accordance with NAVAIR Drawings 3205AS109, 3205AS110, and 3205AS111 shall be subjected to first article testing. Thirty of these DCAs shall be expended in the tests listed in table II, while the remaining DCAs shall be retained for investigative purposes. Any damage inflicted by the environmental exposures that would adversely affect the performance of the item in service application shall be cause for rejection of the first article sample. First article testing is for the purpose of determining that the contractor, prior to starting production, is capable of producing items that comply with the technical requirements as specified in the contract.
As specified (see 6.2), a first article inspection shall be required if two or more years have passed since the item was manufactured, after major turnover in personnel, following major changes to the production facility, or following transfer of production to a new facility or site. Any production prior to notification by the contracting agency of the first article sample acceptability is at the contractor's risk. Failure of any DCA to comply with the requirements of table II shall be cause for rejection of the first article represented.

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TABLE II. First article inspection.

Test sequence	Requirement paragraph	Test paragraph	Sample groups								
			I	II	III	IV	V	VI	VII	VIII	IX
Visual	3.4.1	4.5.4	6	3	3	3	3	3	3	3	3
Radiographic	3.4.2	4.5.5	6	3	3	3	3	3	3	3	3
Leakage	3.4.3	4.5.6	6	3	3	3	3	3	3	3	3
Vibration	3.5.1	4.5.7	6	-	-	-	-	-	-	-	-
Shock	3.5.2	4.5.8	-	3	-	-	-	-	-	-	-
TSHA	3.5.3	4.5.9	-	-	3	-	-	-	-	-	-
Salt fog	3.5.4	4.5.10	-	-	-	3	-	-	-	-	-
6-foot drop	3.5.5	4.5.11	-	-	-	-	3	-	-	-	-
40-foot drop	3.5.6	4.5.12	-	-	-	-	-	3	-	-	-
Function test:	3.7	4.5.13	-	-	-	-	-	-	-	-	-
-65±5 °F			2	1	1	1	1	1	-	3	-
70±5 °F			2	1	1	1	1	1	-	-	3
200±5 °F			2	1	1	1	1	1	3	-	-

4.3 Production inspection. All production DCA manufactured shall be inspected and screened in accordance with the examinations and tests listed in table III. DCAs failing to meet the requirements listed in table III shall be rejected and removed from the lot.

TABLE III. Production inspections.

Item	Test sequence	Requirement paragraph	Test paragraph
All components	Corrosion	3.4.5	4.5.2
DCA cord	Detonation velocity	3.4.4	4.5.1
Loaded cord	Explosive weight	3.4.6	4.5.3
DCA	Visual	3.4.1, 3.7	4.5.4
DCA	Radiographic	3.4.2	4.5.5
DCA	End fittings leakage	3.4.3	4.5.6

4.4 Lot acceptance inspection. Lot acceptance inspection shall consist of the examinations and tests specified in table IV. Failure of any sample DCA to comply with the requirements listed in table IV shall be cause for rejection of the lot represented.

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TABLE IV. Lot acceptance inspection and tests.

Inspection/test	Requirement paragraph	Test paragraph	Quantity
Visual inspection	3.4.1, 3.7	4.5.4	Test and retain sample
Radiographic inspection	3.4.2	4.5.5	Test and retain sample
Leakage	3.4.3	4.5.6	Test and retain sample
Function test: -65±5 °F 70±5 °F 200±5 °F	3.6	4.5.13	 ⅓ sample ⅓ sample ⅓ sample

4.4.1 Sample size. A random sample of DCAs from each production lot, including the samples retained for investigative purposes, shall be selected in accordance with table V for lot acceptance inspection. Test sample DCAs and samples retained for investigative purposes shall not be applied as part of the quantity specified (see 6.2).

TABLE V. Lot acceptance sampling.

Lot size	Test sample size	Retained sample size
2 to 50	9	1
51 to 90	15	1
91 to 150	24	2
151 to 250	27	2
251 to 500	30	3

4.5 Inspections and tests.

4.5.1 Detonation velocity test. To meet the requirements of 3.4.4, a 40-inch length of cord at each end and every 500 feet of planned cord production shall be tested for velocity measurement prior to producing any DCAs.

4.5.2 Examination for corrosion. All parts shall be visually inspected in accordance with 3.4.5 to ensure there are no corrosion products on the hardware and that the threads are coated with dry film lubricant.

4.5.3 Explosive weight test. Every 1,000 feet, a 1-meter section of cord shall be taken and the explosive shall be removed from the cord. The residue shall be weighed to the nearest milligram to verify explosive weight in accordance with 3.4.6. The explosive shall be removed in a safe and approved procedure.

4.5.4 Visual inspection. The external condition and appearance of the DCA shall be determined by comparison to NAVAIR Drawings 3205AS109, 3205AS110, and 3205AS111. Specific inspections of the length, covering, booster tips, seals, and labels shall be conducted at a minimum. Each DCA shall meet the requirements of 3.4.1 and 3.7.

4.5.5 Radiographic examination. Radiographic examination shall be conducted in accordance with ASTM E1742/E1742M. All DCAs shall be identified with serial numbers prior to examination. The DCA shall be arranged on boards or trays in consecutive order with any missing serial numbers identified on the radiographic plate. If these are too long to get on one plate, the ends need to be identified as “A” or “B” ends. Each radiograph shall carry a permanent identification of the items displayed thereon in a region that is 4 x 6 inches maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the lot acceptance tests. Any

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observable imperfections as outlined in 3.4.2.1 and 3.4.2.2 shall be cause for rejection of the DCA. Defective DCA found during radiographic review shall be marked on the radiographic plate and removed from the production lot. Settings on the radiographic equipment shall be noted on the radiographic review sheets.

4.5.5.1 X-ray. X-ray shall show each DCA end assembly and approximately 3 inches of line behind the tips.

4.5.5.2 N-ray. N-ray shall show each DCA end assembly and approximately 3 inches of line behind the tips.

4.5.6 Leakage test. Each production unit shall be leak tested by bombing the DCA with helium at a pressure of 14.7±1.4 psig. The DCA shall then be washed with dry nitrogen and placed in a vacuum chamber and checked for leakage at one atmosphere differential. Certification of the test results and Government witnessing shall be as specified (see 6.2). The DCA shall meet the requirements of 3.4.3.

4.5.7 Vibration. The DCA shall be subjected to the following vibration profiles: two units at -65 °F, two units at 70 °F, and two units at 200 °F. Each DCA shall be subjected to the combined source dwell and random excitation environments or individual source dwell and random excitation environments as applicable at -65, +70, and +200 °F in each axis for 20 minutes. The DCA shall meet the requirements of 3.5.1. The following tolerances shall apply to all vibration testing:

a. Instantaneous random vibration acceleration peaks shall be limited to 3 times the root mean square (RMS) acceleration (3 sigma clipping).

b. The random vibration spectrum shall be within the tolerance range of ±3 dB between 10 and 2,000 Hz. Deviations of ±6 dB are permissible over a cumulative bandwidth of 200 Hz maximum between 500 and 2,000 Hz.

c. Overall RMS acceleration levels shall be within ±1.5 dB of the nominal input level.

d. Sine tones shall be within ±10 percent of the nominal value.

4.5.7.1 Resonance survey. A resonance survey shall be conducted for each DCA configuration in each of the three mutually perpendicular axes, both prior to and following exposure to the source dwell environment. To determine the DCA configuration's resonant frequency and transmissibility, a low level (0.5 g peak), 5 to 500 Hertz (Hz) sweep at a frequency rate of ½ octave per minute (15 minutes duration) with a tolerance equal to ±10 percent on the vibration amplitude and ±2 percent or 1 Hz shall be conducted. The same DCA serial number of each configuration and same mounting location on the vibration fixture shall be utilized for both the pre- and post-source dwell frequency survey. The resonant frequency shall be considered to be any natural frequency with an amplification factor of 2.0 or greater output response relative to the input level. The DCA response shall be measured on the least supported location or a location shown to have significant amplification. All resonance surveys shall be conducted at ambient temperature. Following the completion of the source dwell vibration testing at -65 and 200 °F, the DCA shall be returned to ambient temperature prior to completion of the post-source dwell resonance survey.

4.5.7.2 Source dwell vibration and random vibration excitation. Predicated upon the outcome of the resonance survey, source dwell vibration and random vibration excitation should be conducted simultaneously. If the resonance survey indicates that the four Government-desired test frequencies exceed the permitted test frequencies, the source dwell vibration and random excitation shall not be applied simultaneously and the source dwell vibration levels shall be increased by 0.5 g. Source dwell vibration shall be performed along each of the three mutually perpendicular axes at each of the four amplitudes and frequencies established by the Government based on the resonance survey. Source dwell input levels are shown on figure 1 and shall include the superimposed broad band random background levels of the random excitation as described in 4.5.7.c. Random excitation shall be conducted in accordance with MIL-STD-810, method 514, technique 3. Random excitation shall be a flat spectrum at 0.02 g^2/Hz over the bandwidth from 10-2,000 Hz, as shown on figure 2.

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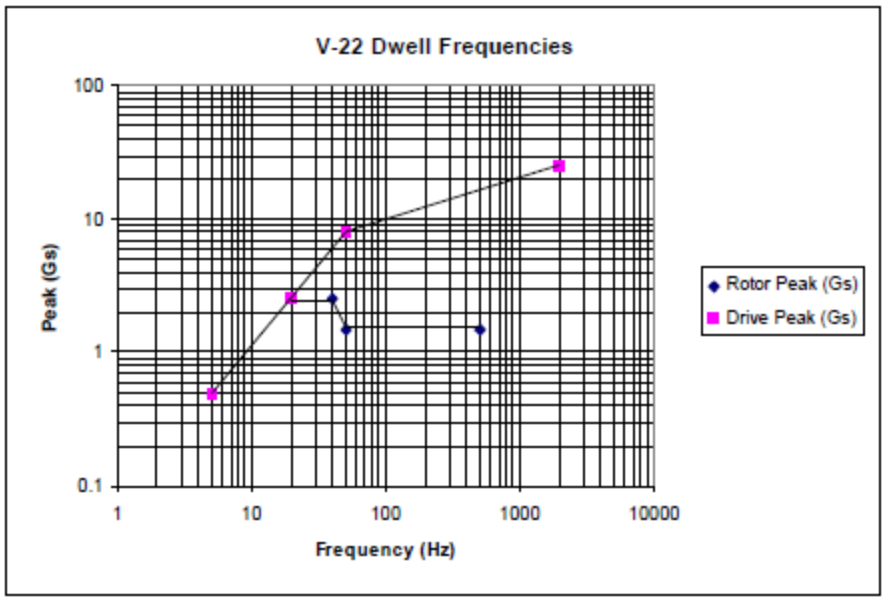


FIGURE 1. Source dwell input levels.

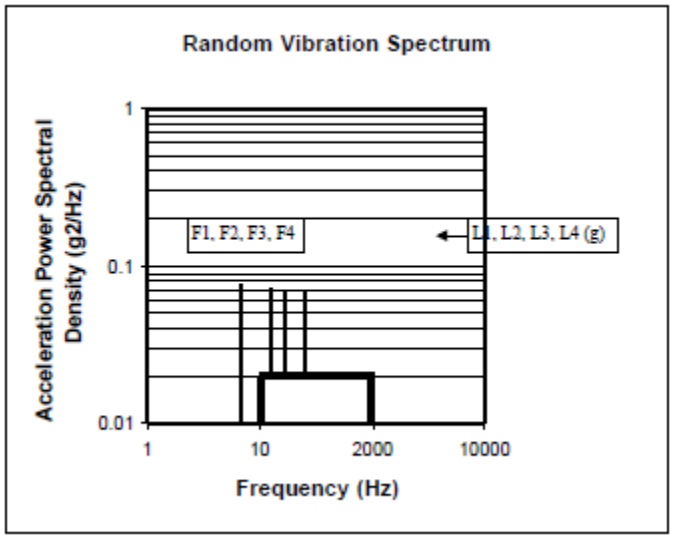


FIGURE 2. Random vibration spectrum.

4.5.8 Shock test. The DCA shall meet the requirements of 3.5.2. The DCA with protective caps (see 6.4.1) installed on the ends shall be subjected to the shock test outlined in MIL-STD-810, method 516, procedure I to the levels on figure 3, three shocks in each direction along each of the three axes for a total of 18 shocks. All shocks will be conducted at ambient temperature.

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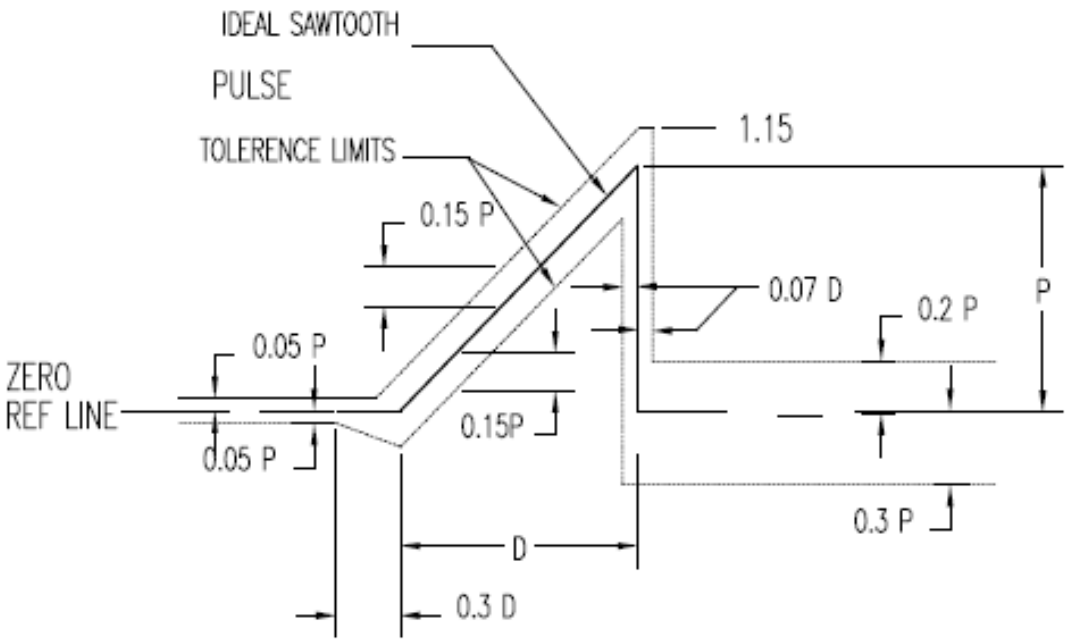


FIGURE 3. Terminal peak sawtooth shock pulse.

4.5.9 TSHA cycling test. The DCA shall meet the requirements of 3.5.3. Temperature recording charts shall be used on all conditioning chambers to verify temperature cycles. The DCA with protective caps installed shall be subjected to the TSHA cycling test outlined in MIL-D-21625 with the following conditions:

a. The DCA shall be supported in such a way that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test. No DCA shall be in contact with another DCA during the cycling schedule shown in table VI.

b. Failure of any DCA to comply with the requirements of 3.5.3 shall be cause for rejection of the lot or first article.

c. Thirty thousand (30,000) feet shall be used in lieu of 70,000 feet for the altitude test.

d. There shall be no interim withdraw of units.

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TABLE VI. TSHA cycling schedule.

DAY	TIME	PROCEDURE
Monday	0800	Place DCA in chamber at 70 °F at 50 percent RH [1/].
	1200	Raise chamber temperature to 160 °F and an RH of 95 percent. The chamber shall reach 160 °F no later than 1300.
	1600	Remove the DCAs from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 30,000 feet (4.36 psi [2/]).
Tuesday	0800	Remove DCAs from chamber and immediately place in chamber maintained at 70 °F at 50 percent RH.
	1200	Remove the DCAs from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 30,000 feet (4.36 psi).
	1600	Remove DCAs from above chamber and immediately place in a chamber maintained at 160 °F at 95 percent RH.
Wednesday	0800	Reduce chamber temperature to 70 °F and RH to 50 percent. The chamber temperature shall reach 70 °F at 50 percent RH no later than 0900.
	1200	Raise chamber temperature to 160 °F and an RH of 95 percent. The chamber shall reach 160 °F no later than 1300.
	1600	Remove the DCAs from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 30,000 feet (4.36 psi).
Thursday	0800	Remove DCAs from chamber and immediately place in chamber maintained at 70 °F at 50 percent RH.
	1200	Remove the DCAs from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 30,000 feet (4.36 psi).
	1600	Remove DCAs from above chamber and immediately place in a chamber maintained at 160 °F at 95 percent RH.
Friday	0800	Reduce chamber temperature to 70 °F and RH to 50 percent. The chamber temperature shall reach 70 °F at 50 percent RH no later than 0900.
	1200	Raise chamber temperature to 160 °F and an RH of 95 percent. The chamber shall reach 160 °F no later than 1300.
	1600	Remove the DCAs from above chamber and immediately place in a chamber maintained at -65 °F at standard ambient pressure.

FOOTNOTES:

1/ Relative humidity.

2/ Pound-force per square inch.

NOTE: This schedule shall be followed for a total of 4 weeks (28 days), except that on the second and fourth weekend, the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of 160 °F at 95 percent RH.

4.5.10 Salt fog test. The DCA, with protective caps installed, shall be subjected to the salt fog test outlined in MIL-STD-810, method 509, procedure I with the following conditions:

a. Waxed string shall be used to support the DCAs in the chamber. The DCAs shall be oriented in an end-up position. No change in orientation is required during the test. The DCAs shall be exposed to a concentration of 5 percent salt corrosive agent for a constant wetting duration of 168 hours at 95 °F (35 °C). Condensed fog shall be collected at two points every 12 hours and the specific gravity, concentration, and pH of the solution measured.

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b. After completing the 168 hours of testing, the salt deposits (if any) shall be removed, if necessary, by a gentle washing with water not warmer than 100 °F.

c. After completing the washing, the units shall be stored for a 48-hour drying period at 70 °F. After 48 hours, a visual inspection shall be conducted. Any signs of damage or degradation shall be noted.

d. The failure of any DCA to comply with the requirements of 3.5.4 shall be cause for rejection of the lot or first article.

4.5.11 Six-foot drop test. The DCAs, with protective shipping caps removed, shall be subjected to the drop test outlined in MIL-D-21625 with the following conditions:

a. The DCAs shall be dropped from a height of 6 feet onto a 2-inch thick steel plate supported by reinforced concrete. One DCA shall be used for each drop orientation. All impact orientations are given on figure 4.

b. After each DCA is dropped, a visual inspection shall be conducted to document any external damage that occurred and to verify non-initiation of the DCA.

c. If it is determined that the damage sustained by any unit caused it to be unsafe for handling, all testing shall be stopped.

d. Failure of any DCA to comply with the requirements of 3.5.5 shall be cause for rejection of the lot or first article.

4.5.12 Forty-foot drop test. The DCAs, with protective caps removed, shall be subjected to the 40-foot test outlined in MIL-STD-331, test 103, with the following conditions:

a. The DCAs shall be dropped 40 feet onto a 2-inch thick steel plate supported by reinforced concrete. One new DCA shall be used for each drop orientation. All impact orientations are given on figure 4.

b. After each drop, a visual inspection shall be conducted to document any external damage that occurred and to verify non-initiation of the DCA.

c. If it is determined that the damage sustained to a DCA causes it to be unsafe for handling, all testing shall be stopped.

d. Failure of any DCA to comply with the requirements of 3.5.6 shall be cause for rejection of the lot or first article.

e. After the visual inspection, each DCA shall then be disposed of in a safe and approved manner in accordance with local, state, and Federal regulations.

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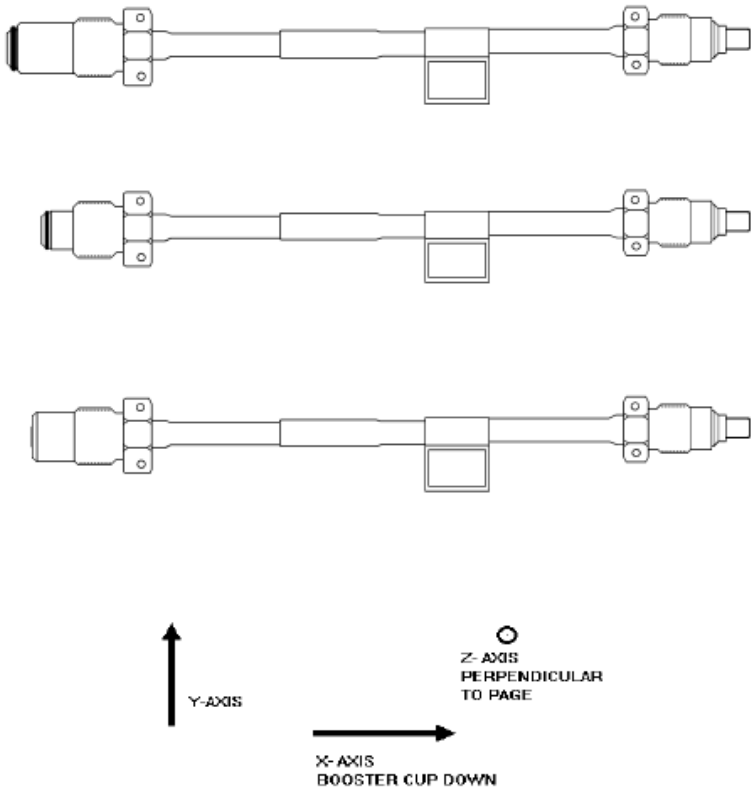


FIGURE 4. Shock/vibration and drop axis.

4.5.13 Functional performance tests. Sample DCAs for the functional performance test firings shall be conditioned for not less than 2 hours and not more than 24 hours at the specified temperature. The DCAs shall be removed from the conditioning chamber and fired in the appropriate test fixture within 5 minutes. If any DCA is not fired within the 5-minute time period after removal, it shall be reconditioned at the specified temperature for an additional 2 hours and then tested. The DCA shall meet the performance requirements of 3.6.

4.5.13.1 Primer-high. The firing fixture (NAVAIR Drawing 3205AS182) shall be used to function the percussion primer in the end of the DCA. The DCA shall not vent past 2 inches from the crimp at either end. The indentation in the dent block shall be measured from three reference location. The output performance shall meet the requirements of 3.6.1. The test firing set up is shown on figure 5.

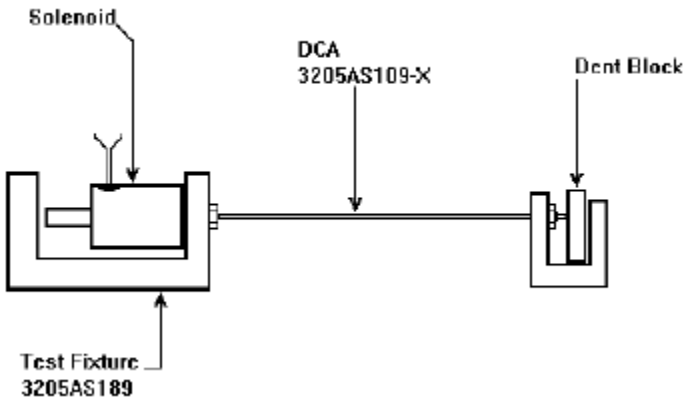


FIGURE 5. Primer-high test setup.

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4.5.13.2 High-low. The DCA shall accept a high order detonation signal from an explosive test tip and propagate the complete length. The DCA shall not vent within 2 inches of an end. The low energy end shall not be expelled from the connector. The ballistic output of the high energy end shall meet the requirements of 3.6.2. The low energy end shall not function a high energy end. The test firing set up is shown on figure 6.

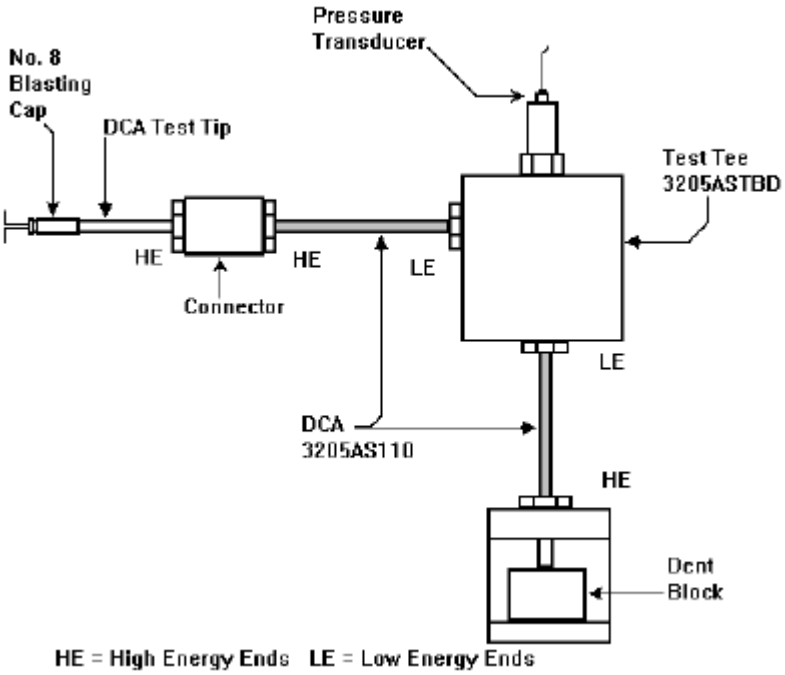


FIGURE 6. High-low test setup.

4.5.13.3 High-gas. The closed bomb shall be maintained at 70±5 °F for all firings. The closed bomb shall be cleaned after each firing in accordance with the standard operating procedure. The closed bomb volume shall be measured and recorded prior to the start of each lot acceptance test and at the conclusion of the lot acceptance test. The test firing setup is shown on figure 7. The DCA shall not vent past 2 inches from either end. The ballistic pressure shall meet the requirements of 3.6.3.

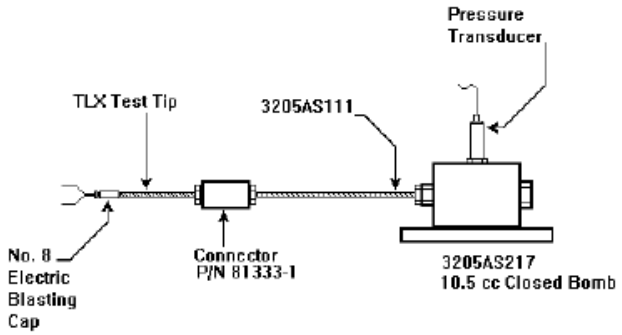


FIGURE 7. High-gas test setup.

4.5.14 Test failure. If a test failure is attributable to an assigned cause, excluding the DCA, the test shall be declared a no-test and repeated.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These DCAs were developed under Bell-Boeing specification 901-947-784 for use in the Emergency Egress Hatch Removal System (EEHRS) on V-22 Aircraft. They are components of the EEHRSs installed in the MV-22, CV-22 and HV-22 Aircraft. They are the ballistic signal transmission components of the system. The system is designed to provide emergency egress in a post-crash environment on land or water. The primer-high DCA is used to accept mechanical energy from the initiators and convert it to explosive energy. The high-low DCA acts as a one-way valve in the seat drop function. The high-gas DCA provides the gas pressure to function the seat drop thruster. No changes to the configuration are permitted Since the EEHRSs are designed for use only in military aircraft, there are no commercial applications for the DCA.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1 and 4.2).

c. Materials required, as established by the technical service release (see 3.3).

d. First article sample size, if other than as specified (see 4.2).

e. Production lot size and test facility (see 4.3).

f. Lot acceptance ballistic sample size, if other than as specified (see 4.4.1).

g. Packaging requirements (see 5.1).

h. Applicable safety precaution requirements (see 6.5).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to bidders whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1) and indicate the number of items to be tested in accordance with 4.2. The contracting officer should also include specific instructions in acquisition documents of the first articles. Invitation(s) for bid(s) should provide that the U.S. Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the U.S. Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior U.S. Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Caps.

6.4.1.1 Aluminum witness caps. Aluminum witness caps are installed on the high explosive booster cup end.

6.4.1.2 Plastic shipping caps. Plastic shipping caps are installed on the other end of DCA.

6.4.2 DCA. DCAs are the explosive transmission mechanism used to provide a transfer path of explosive stimulus from one pyrotechnic device to another.

6.4.2.1 Gas generator-equipped DCA. The gas generator-equipped DCA provides pressure to function a thruster, converting the ballistic stimulus into mechanical work.

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6.4.2.2 High explosive/low energy DCA. The high explosive/low energy DCA acts as a one-way valve to explosive stimulus.

6.4.2.3 Percussion primer-equipped DCA. The percussion primer-equipped DCA accepts mechanical energy input and transitions it into an explosive stimulus.

6.4.3 Flight critical part. A part, the single failure of which, during any operating condition, could cause the loss of aircraft or one of its major components, loss of control, or significant personnel injury, including during flight. These DCAs are considered flight critical parts and the technical data package should be identified and procured as such.

6.4.4 Primary component. Primary components are all components in which a functional failure would result in a misfire or malfunction of the DCA.

6.5 Safety precautions. When specified (see 6.2), the safety precaution requirements of DoD Manual 4145.26, "DoD Contractors' Safety Manual for Ammunition and Explosives," are applicable. NOTE: When this specification is used as a part of the description of work to be accomplished by a Government activity, the safety precaution requirements of NAVSEA OP 5, "Ammunition and Explosives, Safety Ashore," are applicable.

6.6 Subject term (key word) listing.

Aluminum
Booster cups
Gas generator
HMX - Homocyclonite
HNS - Hexanitrostilbene
Percussion primer
Thin Layer Explosive (TLX)

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2026-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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